

Mathematics, 2017 EC, entrance exam, version 3

Grade 12 · Mathematics · 2017 E.C.

Time allowed: 120 minutes

Questions: 51

Total marks: 51

Name _____ Grade _____ ID No. _____ Date _____

1. In the set of natural numbers, which one of the following defines a prime number? (1)
 - A. An odd number that has more than two different divisors.
 - B. A number that has exactly three different factors.
 - C. A number that has no positive divisors other than 1 and itself.
 - D. A number that has exactly one divisor.
2. If the $\text{GCF}(a, 169) = 13$ and $\text{LCM}(a, 169) = 338$, what is the value of a ? (1)
 - A. 26
 - B. 394
 - C. 52
 - D. 20
3. If $a = 5 + 2\sqrt{6}$ and $b = 1/a$, then which one of the following is the value of $a^2 + b^2$? (1)
 - A. $98 - 40\sqrt{6}$
 - B. 98
 - C. 1
 - D. $40\sqrt{6}$
4. After rationalizing the denominator of the expression $(3\sqrt{2} + \sqrt{3})/(3\sqrt{2} - 2\sqrt{3})$, what will be the simplified expression? (1)
 - A. $(14+3\sqrt{6})/2$
 - B. $(24+8\sqrt{6})/6$
 - C. $(8+3\sqrt{6})/2$
 - D. $(8-3\sqrt{6})/2$
5. The solution set of the equation $|7 - (x+2)| + |3x-3| = 0$ is equal to: (1)
 - A. $\{-1, 2\}$
 - B. $\{\}$
 - C. $\{-7/4, 1/2\}$
 - D. $\{1/2, 7/4\}$
6. Let A, B and C be three sets: $A = \{\text{natural numbers less than } 5\}$, $B = \{x: x \in \mathbb{N} \text{ and } x \text{ is a factor of } 6\}$, $C = \{3, 4, 5, 6\}$. Which of the following is true? (1)
 - A. $A \cap B = A$
 - B. $A \cup (B \cap C) = B$
 - C. $A \cup B = B$
 - D. $A \cap (B \cup C) = A$
7. A line from the top of a cliff to the ground passes over the top of a pole 20m high. The line meets the ground at a point 15m from the base of the pole. If it is 120m away from this point to the base of the cliff, how high is the cliff? (1)
 - A. 160
 - B. 90
 - C. 320
 - D. 180
8. Which one of the following is the mean deviation about the mean of the data: 38, 44, 70, 54, 40, 46, 63, 55, 42, 48? (1)
 - A. 50
 - B. 5
 - C. 8.4
 - D. 8

9. The following table shows an analysis of monthly wages for two firms A and B. Which statement is true? (Mean wages same 5253, variance A=100, B=121) (1)
- A. Firm A pays larger amount of monthly wages.
B. Firm B has greater variability in the individual wages.
C. Firm A has greater variability in the individual wages.
D. Firm B pays smaller amount of monthly wages.
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10. If the mean of the following data is 28, find missing frequency f. (Class intervals 0-10,10-20,20-30,30-40,40-50,50-60; frequencies 12,18,27,f,17,6) (1)
- A. 14
B. 4
C. 20
D. 10
-
11. Which one of the following numbers is the coefficient of a^5b^7 in expansion of $(a - b)^{12}$? (1)
- A. 101376
B. 792
C. -101376
D. -792
-
12. If $2/6! + 1/7! = x/8!$, then $x = ?$ (1)
- A. 120
B. 26.13
C. 56
D. 840
-
13. Number of ways of selecting 9 balls from 6 red, 5 white, 5 blue if each selection consists of 3 balls of each colour? (1)
- A. 40
B. 2000
C. 48000
D. 160
-
14. A man cuts three lengths from a board of length 91cm. Second length is 3cm longer than shortest, third is twice shortest. The third piece is at least 5cm longer than second. Find possible lengths of shortest piece. (1)
- A. $[0,8)$
B. $(0,8)$
C. $[8,22]$
D. $(22,\infty)$
-
15. Let $A=\{1,2,\dots,20\}$. Relation R from A to A defined by $R=\{(x,y): 3x-y=0\}$ where $x,y\in A$. Find the range of R. (1)
- A. $\{1,2,3,4,5,6\}$
B. $\{3,6,9,12,15,18\}$
C. $\{6,9,12,15,18,21\}$
D. $\mathbb{R}$
-
16. A question paper has 12 questions: part I (5 questions), part II (7 questions). Student must attempt 8 questions, at least 3 from each part. Number of ways? (1)
- A. 210
B. 175
C. 385
D. 420
-
17. Sum of all natural numbers between 100 and 1000 exclusive, which are divisible by 5. (1)
- A. 99550
B. 97350
C. 550000
D. 98450
-
18. Sum of some terms of a geometric sequence is 315, first term 5, common ratio 2. Find last term. (1)
- A. 160
B. 180
C. 224
D. 150

19. Given a graph of system of linear inequalities (image). Which system is represented by the shaded region? (1)
- A. $y \leq x+2, y \leq -x+2, y \geq 0$ B. $y \geq x+2, y \geq -x+2, y \leq 0$
C. $y \leq x+2, y \geq -x+2, y \geq 0$ D. $y \geq x+2, y \leq -x+2, y \leq 0$
-
20. 12th term of geometric sequence whose 8th term is 192 and common ratio 2. (1)
- A. 3072 B. 6144
C. 4450 D. 9216
-
21. Sum to n of the series $1/(1 \times 2) + 1/(2 \times 3) + 1/(3 \times 4) + \dots$ (1)
- A. $(n+1)/n$ B. $1/(n(n+1))$
C. $n/(n+1)$ D. $n(n+1)$
-
22. Sum of squares of two consecutive odd positive integers is 290. Find the numbers. (1)
- A. -13 and 11 B. 13 and 15
C. 11 and 13 D. 9 and 11
-
23. Which sampling method involves dividing population into subgroups (strata) then randomly selecting from each stratum? (1)
- A. Multistage sampling B. Systematic sampling
C. Cluster sampling D. Stratified sampling
-
24. If A and B are 3×3 matrices with $\det(A)=6, \det(B)=5$, which statement is true? (1)
- A. $\det(3A^T B)=90$ B. $\det(2AB)=240$
C. $\det(4A)=24$ D. $\det(2B^2)=50$
-
25. Slope of tangent line to $f(x)=x^3-3x+2$ at $x=3$. (1)
- A. 20 B. $-1/24$
C. 24 D. 26
-
26. Volume of cube increasing at $8 \text{ cm}^3/\text{s}$. How fast is surface area increasing when edge is 12 cm? (1)
- A. $2/9 \text{ cm}^2/\text{s}$ B. $8/3 \text{ cm}^2/\text{s}$
C. $3 \text{ cm}^2/\text{s}$ D. $12 \text{ cm}^2/\text{s}$
-
27. Marginal revenue when $x=10$ for $R(x)=13x^2+26x+15$. (1)
- A. 286 B. 1575
C. 301 D. 26
-
28. Largest interval on which $f(x)=2x^3-3x^2-36x+7$ is increasing. (1)
- A. $(-\infty, 2] \cup [3, \infty)$ B. $(-\infty, -3] \cup [2, \infty)$
C. $(-\infty, -2] \cup [3, \infty)$ D. $[-2, 3]$
-
29. Area bounded by $y=x^3$ and x-axis between $x=-2$ and $x=1$. (1)
- A. $16/4$ B. $15/4$
C. 9 D. $17/4$
-
30. Local maximum value of $f(x)=x^3-6x^2+9x+15$. (1)
- A. $f(1)$ B. $f(3)$
C. $f(-1)$ D. $f(-3)$

31. Area of parallelogram with adjacent sides $i+k$ and $2i+j+k$. (1)
- A. $\sqrt{2}$ B. $\sqrt{3}$
C. 2 D. 3
-
32. Coordinates of centroid of triangle with vertices A(2,1), B(5,8), C(8,3). (1)
- A. (4,5) B. (5,8)
C. (5,2) D. (5,4)
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33. Sequence defined by $a_n = a_{n-1} + (a_{n-2})^2 + n$, with $a_1=1$, $a_2=-2$. Find a_3 ? (Options: 29,175,125,493) Likely a_3 ? (1)
Actually question says 'value of a_n ' but options are numbers; probably a_3 or a_4 ? Let's compute: $a_3 = a_2 + (a_1)^2 + 3 = -2+1+3=2$. Not in options. $a_4 = a_3 + (a_2)^2 + 4 = 2+4+4=10$. Not. $a_5 = a_4 + (a_3)^2 + 5 = 10+4+5=19$. Not. $a_6 = 19 + (10)^2 + 6 = 19+100+6=125$. So $a_6=125$. Answer C.
- A. 29 B. 175
C. 125 D. 493
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34. Partial fraction decomposition of $(x^2+3)/(x^3+x^2+x+1)$. (1)
- A. $2/(x+1) - (1+x)/(x^2+1)$ B. $2/(x+1) + (1-x)/(x^2+1)$
C. $1/(x+1) + (x-1)/(x^2+1)$ D. $1/(x+1) - (x+1)/(x^2+1)$
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35. Solution set of $(3x)/(x+4) + 2/(x^2+6x+8) = 1/(x+2)$. (1)
- A. $\{-2\}$ B. $\{-2, 1/3\}$
C. $\{\}$ D. $\{1/3\}$
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36. Solution set of system: $x+y+z=1$, $y+z=3$, $x-y+z=-5$? (The PDF has garbled text, but likely this). (1)
- A. $\{(-2,2,1)\}$ B. $\{(2,-2,1)\}$
C. $\{(2,1,-2)\}$ D. $\{(-2,-2,1)\}$
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37. Which is NOT true about graph of $f(x)=(x+3)/(x^2+5x+6)$? (1)
- A. Graph has hole at $(-3,-1)$ B. Horizontal asymptote is x-axis
C. Vertical asymptotes at $x=-3$ and $x=-2$ D. Graph has no x-intercept
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38. Image of point (1,-3) under translation T if image of (-1,2) is (3,-1). (1)
- A. (4,-3) B. (3,0)
C. (5,-6) D. (5,6)
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39. Find fifth decile (D_5) of data: 10,13,15,12,18,15,16,12,13,20,17,14,19,18,23,2. (1)
- A. 15.5 B. 16
C. 12.5 D. 15
-
40. If $u = 3i+2j+5k$ and $v = -3i+2j+k$, then $(3u) \cdot (u \times v) = ?$ (1)
- A. -12 B. 12
C. 0 D. -6
-
41. Which series is divergent? (1)
- A. $\sum [(-1)^{n+1} * (2/3)^n]$ B. $\sum 1/(2n-1)$
C. $\sum [(2/4)^n + (3/5)^n]$ D. $\sum [(2/3)^{n-1} * (1/2)^{n-1}]$

42. $\int_0^2 (3x^2+2x) dx = ?$ (1)
A. 20 B. 10
C. 12 D. 8
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43. Toss a coin four times. Number of outcomes with at least two tails. (1)
A. 9 B. 12
C. 11 D. 8
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44. If $f(x)=\sqrt{x^2+2x+1}$, then $f'(2)=?$ (1)
A. 2 B. $1/3$
C. $2/3$ D. 1
-
45. Sum of infinite series $0.33 + 0.0033 + 0.00033 + \dots$ (1)
A. $33/(10^n)$ B. $2/3$
C. $33/(10^{(n+1)})$ D. $1/3$
-
46. Box with 8 items, 6 defective. Select 3 items. Probability that two defective and one non-defective. (1)
A. $3/14$ B. $15/56$
C. $5/56$ D. $15/28$
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47. Which relation is a function? (1)
A. $R=\{(1,0),(4,5),(1,3),(6,-6)\}$ B. $R=\{(x,y): x,y \in \mathbb{R}, x+y=9\}$
C. $R=\{(x,y): x,y \in \mathbb{R}, x \text{ is father of } y\}$ D. None
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48. Which matrix is in reduced echelon form? (1)
A. $[[1,0,0],[0,1,0],[0,0,0]]$ B. $[[1,2,0],[0,0,1],[0,0,0]]$
C. $[[1,0,2],[0,1,-1],[0,0,0]]$ D. All of the above
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49. Vector equation of line $3x+5y=15$. (1)
A. $r = 3j + \lambda(3i+5j)$ B. $r = 5i + \lambda(5i-3j)$
C. $r = 3i + \lambda(3i-5j)$ D. $r = 5j + \lambda(5i+3j)$
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50. Which statement is NOT true about reflection? (1)
A. Every point moves same direction same distance
B. Distance is preserved
C. One-to-one mapping
D. Reflection in line L is a function
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51. Which statement is NOT true? (1)
A. In normal distributions, Pearson's skewness = 0
B. In symmetrical distribution, Pearson's skewness < Bowley's
C. If mode < median < mean, positively skewed
D. If mean < median < mode, Bowley's skewness $\beta < 0$

— END OF EXAMINATION —